

CS-C3240 Machine Learning

Predicting Finland’s electricity consumption from the weather and time

Project report

1 Introduction

Weather affects how people use electricity. Cold weather increases demand for heating, while hot weather causes air conditioners to run longer. Strong winds draw heat away from structures more quickly, further increasing heating demand. High humidity makes buildings feel hotter than they actually are, increasing demand for air conditioning. Low humidity, on the other hand, makes buildings feel colder, increasing demand for heating [1]. Electricity consumption also follows daily and weekly patterns independent of weather, since usage tends to vary by hour of the day and by day of the week [2].

The goal of this project is to predict how weather conditions in Finland, namely temperature, wind, and humidity, along with the hour of day, weekday and month, affect electricity consumption. This is done by combining data from Fingrid, the Finnish electricity transmission system operator, and the Finnish Meteorological Institute (FMI). This report consists of an introduction, problem formulation, methods, results, conclusion, and references. The methods section further consists of data preparation, model choice and model validation.

2 Problem formulation

The data in this study comes from Fingrid and the Finnish Meteorological Institute (FMI). The electricity data is the “Electricity consumption in Finland” dataset from Fingrid’s open data service [3], comprising 119 369 measurements, and the weather data consists of 52 589 hourly observations from the FMI Tapiola weather station, downloaded from FMI’s open data service [4]. Both cover the years 2020–2025.

Each data point represents one hour and has seven numerical features: air temperature in degrees Celsius (`temp_c`), wind speed in metres per second (`wind_ms`), relative humidity as a percentage (`humid_pc`), a cyclic encoding of the hour of day as sine and cosine (`hour_sin`, `hour_cos`), the weekday as an integer from 1 to 7 (`weekday`), and the month as an integer from 1 to 12 (`month`). The hour of day is encoded cyclically so that hour 23 and hour 0 are treated as adjacent rather than maximally distant. The label is the average electricity consumption in Finland during that hour, measured in megawatt-hours (`consumption_mwh`). Since the label is continuous, this is a supervised regression problem. The first rows of the merged data are shown in Table 1.

time_utc	local_time	consumption_mwh	temp_c	wind_ms	humid_pc	hour	hour_sin	hour_cos	weekday	is_weekend	month
2019-12-31T22:00:00Z	2020-01-01 00:00:00	9822	-1.5	4.3	83	0	0.0000	1.0000	3	0	1
2019-12-31T23:00:00Z	2020-01-01 01:00:00	9548	-1.1	4.3	85	1	0.2588	0.9659	3	0	1
2020-01-01T01:00:00Z	2020-01-01 03:00:00	9141	-0.7	4.5	87	3	0.7071	0.7071	3	0	1
2020-01-01T02:00:00Z	2020-01-01 04:00:00	9149	0.1	5.4	83	4	0.8660	0.5000	3	0	1
2020-01-01T03:00:00Z	2020-01-01 05:00:00	9206	0.5	5.2	82	5	0.9659	0.2588	3	0	1
2020-01-01T04:00:00Z	2020-01-01 06:00:00	9432	0.3	4.7	83	6	1.0000	0.0000	3	0	1
2020-01-01T05:00:00Z	2020-01-01 07:00:00	9419	0.1	4.7	85	7	0.9659	-0.2588	3	0	1
2020-01-01T06:00:00Z	2020-01-01 08:00:00	9558	0.8	4.9	81	8	0.8660	-0.5000	3	0	1
2020-01-01T07:00:00Z	2020-01-01 09:00:00	9522	1.5	4.6	77	9	0.7071	-0.7071	3	0	1

Table 1: The first rows of the merged raw data (`hour_sin` and `hour_cos` rounded to four decimals).

3 Methods

3.1 Preparing the data

The dataset is composed of 52 241 data points. It is merged from two datasets: Fingrid’s national electricity consumption data and FMI’s weather data. Fingrid reported consumption hourly before 13 June 2023 and every 15 minutes thereafter, so the Fingrid data was first aggregated to hourly resolution by averaging all measurements within each hour. The two datasets were then merged on the hourly timestamp. 250 hours were discarded because they lacked temperature, wind speed or humidity observations, which gives the final 52 241 data points.

We selected temperature, wind, and humidity based on known physical effects on heating and cooling demand, and included hour of day, weekday, and month to capture daily and seasonal consumption patterns. We decided to drop the columns `time_utc`, `local_time`, `hour`, and `is_weekend`. Columns `time_utc`, `local_time`, and `hour` are redundant once cyclically encoded as `hour_sin` and `hour_cos`, and raw timestamp values are not directly usable as model features. Column `is_weekend` is redundant given that `weekday` already captures the same information. Figures 1–6 show how consumption varies with each of the selected features.

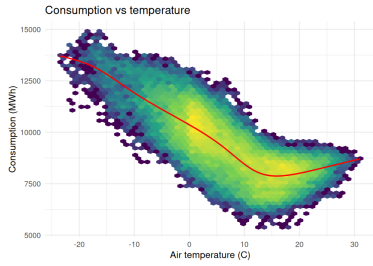


Figure 1: Consumption against air temperature.

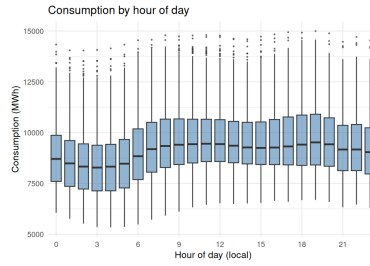


Figure 2: Consumption by hour of day.

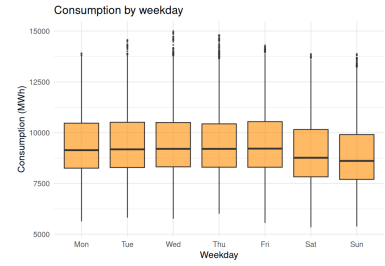


Figure 3: Consumption by weekday.

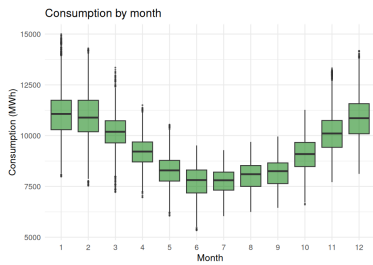


Figure 4: Consumption by month.

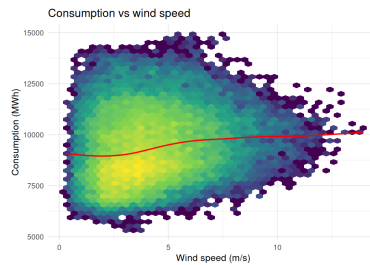


Figure 5: Consumption against wind speed.

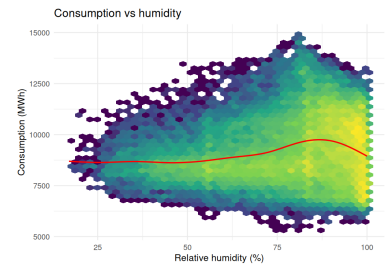


Figure 6: Consumption against relative humidity.

Weekday and month were kept as plain integers rather than cyclically encoded. Unlike hour, where the wraparound between 23:00 and 00:00 occurs within a single day and directly affects short-term consumption patterns, the wraparound effects for weekday and month are less pronounced for this application. The date is only used after processing to sort the input data chronologically for comparison.

3.2 Model

We decided to use a linear regression model for this project, with a hypothesis space consisting of all linear functions of the input features `temp_c`, `wind_ms`, `humid_pc`, `hour_sin`, `hour_cos`, `weekday`, and `month`, together with the squared temperature (see below). This decision was made because the label we are trying to predict is continuous. Linear regression is also a simple baseline that trains instantly on 52 000 rows, and its coefficients are directly interpretable.

Since our dataset consists of data points with numeric labels, this is a regression problem, and we chose squared error loss as our loss function. Squared error loss penalizes large prediction errors much more than small ones. This is useful for our problem because large errors in predicted electricity consumption could cause grid operators to over- or under-supply electricity, which is more costly than a small prediction error.

The correlation matrix (Figure 7) shows a strong linear relationship between temperature and consumption ($r = -0.77$), which supports the use of linear predictors. The scatter plot in Figure 1 nevertheless shows that this relationship flattens above approximately 15°C, so a squared temperature term was added to the hypothesis space. The predictor remains linear in

its parameters and is therefore still a linear regression model:

$$h(\mathbf{x}) = w_0 + w_1x_{\text{temp}} + w_2x_{\text{temp}}^2 + w_3x_{\text{wind}} + w_4x_{\text{humid}} + w_5x_{\text{hour_sin}} + w_6x_{\text{hour_cos}} + w_7x_{\text{weekday}} + w_8x_{\text{month}}. \quad (1)$$

The weights are learned by minimising the squared error loss

$$L((\mathbf{x}, y), h) = (y - h(\mathbf{x}))^2, \quad (2)$$

where y is the measured consumption and $h(\mathbf{x})$ the predicted consumption of a data point.



Figure 7: Pearson correlation matrix of the features and the label.

3.3 Model validation

For model validation the dataset was split chronologically by calendar year. The years 2020–2023 were used for training (34 717 data points, 66.5%), 2024 for validation (8 778 data points, 16.8%) and 2025 for testing (8 746 data points, 16.7%).

We chose this chronological split instead of a random one as we intend to use the model to predict future consumption based on the weather and time data. Thus the validation and test sets must represent time periods the model has not seen during training. Second, consumption in consecutive hours is highly correlated, so a random split would place nearly identical data points in both the training and validation sets. Thus it could give an over-optimistic estimate of its performance when the model interpolates between hours.

Splitting on whole calendar years also ensures that each set covers a complete annual cycle. Had the split been made at an arbitrary point in time, the test set might have contained only summer or only winter months, and the reported error would have reflected a single season rather than year-round performance.

4 Use of AI

We used AI to help write the R scripts to draw the plots. AI was also used to fix grammar and format the LaTeX document.

References

- [1] Edge Branding. Weather's effect on energy use behaviour. Montel.energy, 2024. URL <https://montel.energy/resources/blog/how-does-weather-affect-energy-consumption-behaviour>. Accessed: Sept. 16, 2026.
- [2] U.S. Energy Information Administration (EIA). Electricity demand changes in predictable patterns. Today in Energy, 2011. URL <https://www.eia.gov/todayinenergy/detail.php?id=4190>. Accessed: Sept. 16, 2026.
- [3] Fingrid Oyj. Electricity consumption in Finland. Fingrid Open Data, 2026. URL <https://data.fingrid.fi/en/data?datasets=124>. Accessed: Sept. 16, 2026.
- [4] Finnish Meteorological Institute. Download observations. FMI Open Data, 2026. URL <https://en.ilmatieteenlaitos.fi/download-observations>. Accessed: Sept. 16, 2026.

Appendix

Code: <https://github.com/Erkkuleo/ML-project-predicting-consumption-from-weather>